

Alberta Buck - Identity - Cryptography Example (v1.0)

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This document has two parts. First, a data flow analysis shows where each piece of identity data resides – Ethereum contract storage, Holochain DHT, or local wallet – at each step of the protocol, using mermaid sequence diagrams to trace registration, approval, transfer, and regulatory compliance flows. Second, a complete worked cryptographic example implements every step in Python on the `alt_bn128` (BN254) curve via `py_ecc`, with counter-examples demonstrating what each verification check looks like when it fails.

The scenario is a transfer from Private Alice to Public Bob (an AMM pool). Alice obtains a Pointcheval-Sanders signed identity from an issuer, derives a fresh unlinkable credential via the Identity Fountain, registers it on-chain (~235K gas NIZK verification), and re-encrypts her identity for Bob with a Chaum-Pedersen proof of correct re-encryption (~29K gas). Bob verifies Alice's identity using the two-channel design: the on-chain ElGamal ciphertext provides cryptographic binding, while Holochain delivers the plaintext `identity_data`. The AMM pool scenario demonstrates how this proof chain protects Bob under subpoena – even 18 months and 500,000 transactions later – and why Alice cannot hide from regulatory recovery despite the Identity Fountain's unlinkability guarantees. (PDF, Text)

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1 Data Flow: Where Everything Lives

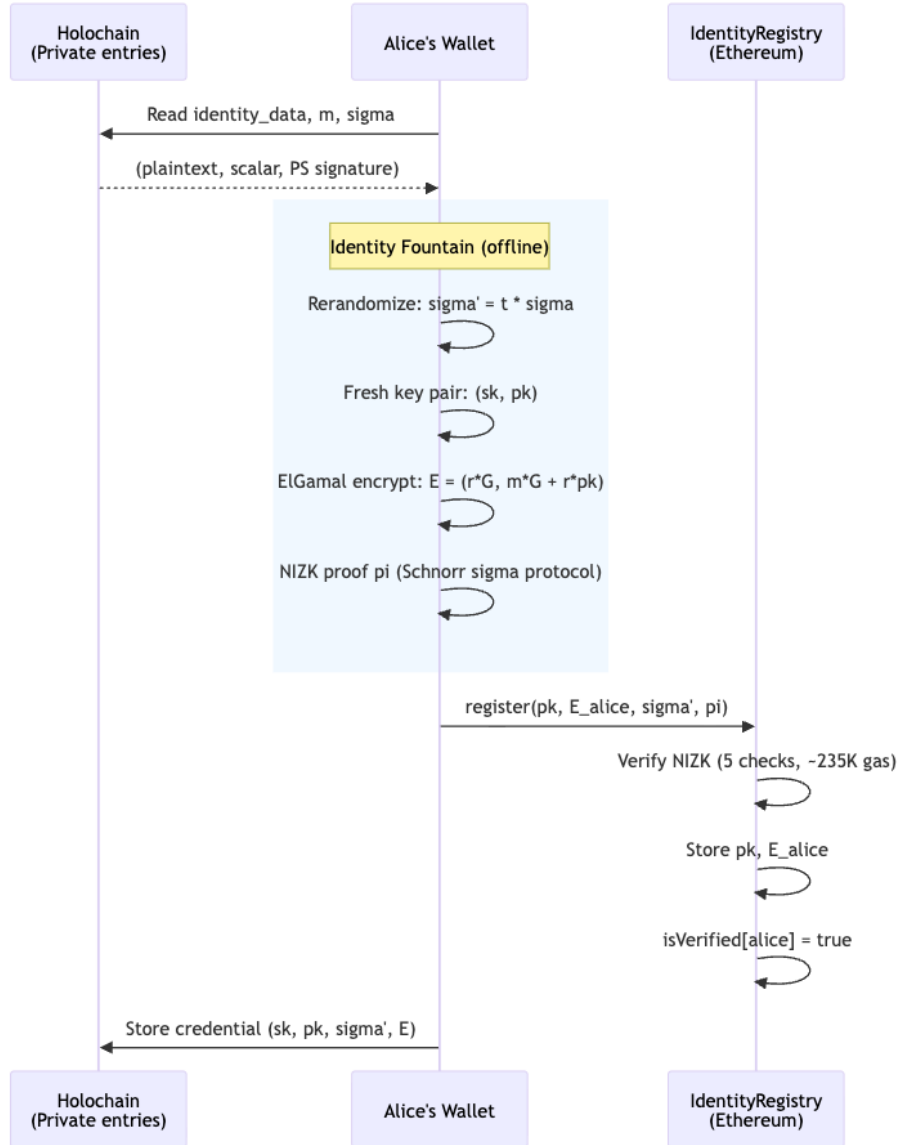
The Alberta Buck identity system spans two networks (Ethereum and Holochain) and a local device (the wallet). The tables and mermaid sequence diagrams below map every piece of identity data to its storage location, visibility, and the protocol step that consumes it – tracing flows through registration, approval, transfer, and two-channel verification, including the trust checkpoint where wallets detect malfeasance, and the AMM pool scenario where a public pool operator responds to a regulatory subpoena months after a swap. The key result: all cryptographic proofs are permanently on-chain, so regulatory compliance requires only Bob’s secret key and an archival Ethereum node – not Alice’s cooperation.

1.1 Data Residency

Data	Location	Visibility	When Used
<code>identity_data</code> (plaintext)	Holochain Private entry	Owner only	Approve (sent to counterparty)
<code>m = H(identity_data)</code>	Holochain Private entry	Owner only	Registration, Approve
PS signature <code>sigma</code>	Holochain Private entry	Owner only	Registration (rerandomized)
Secret key <code>sk</code>	Holochain Private entry	Owner only	Approve (decrypt + re-encrypt)
Rerandomized <code>sigma'</code>	IdentityRegistry	Public (on-chain)	Registration verification
Public key <code>pk</code>	IdentityRegistry	Public (on-chain)	Approve, Transfer
ElGamal ciphertext <code>E_alice</code>	IdentityRegistry	Public (on-chain)	Approve (read by contract)
NIZK proof <code>pi</code>	IdentityRegistry (tx)	Public (on-chain)	Registration verification
Re-encrypted <code>E_bob</code>	BUCK ERC-20 (approve)	Public (on-chain)	Two-channel verification
Chaum-Pedersen proof	BUCK ERC-20 (approve tx)	Public (on-chain)	Approve verification
<code>_receiptFragments[a][b]</code>	BUCK ERC-20 storage	Public (on-chain)	Transfer (bilateral check)
<code>isPublic</code> flag	IdentityRegistry	Public (on-chain)	Transfer (exemption check)
Public <code>identity_data</code>	IdentityRegistry or DHT	Public	Anyone (no approve needed)
Holochain Warrants	Holochain DHT	Public	Malfeasance detection
Canonical receipt	Holochain Private entry	Each party only	Dispute resolution

1.2 Registration: Credential Goes On-Chain

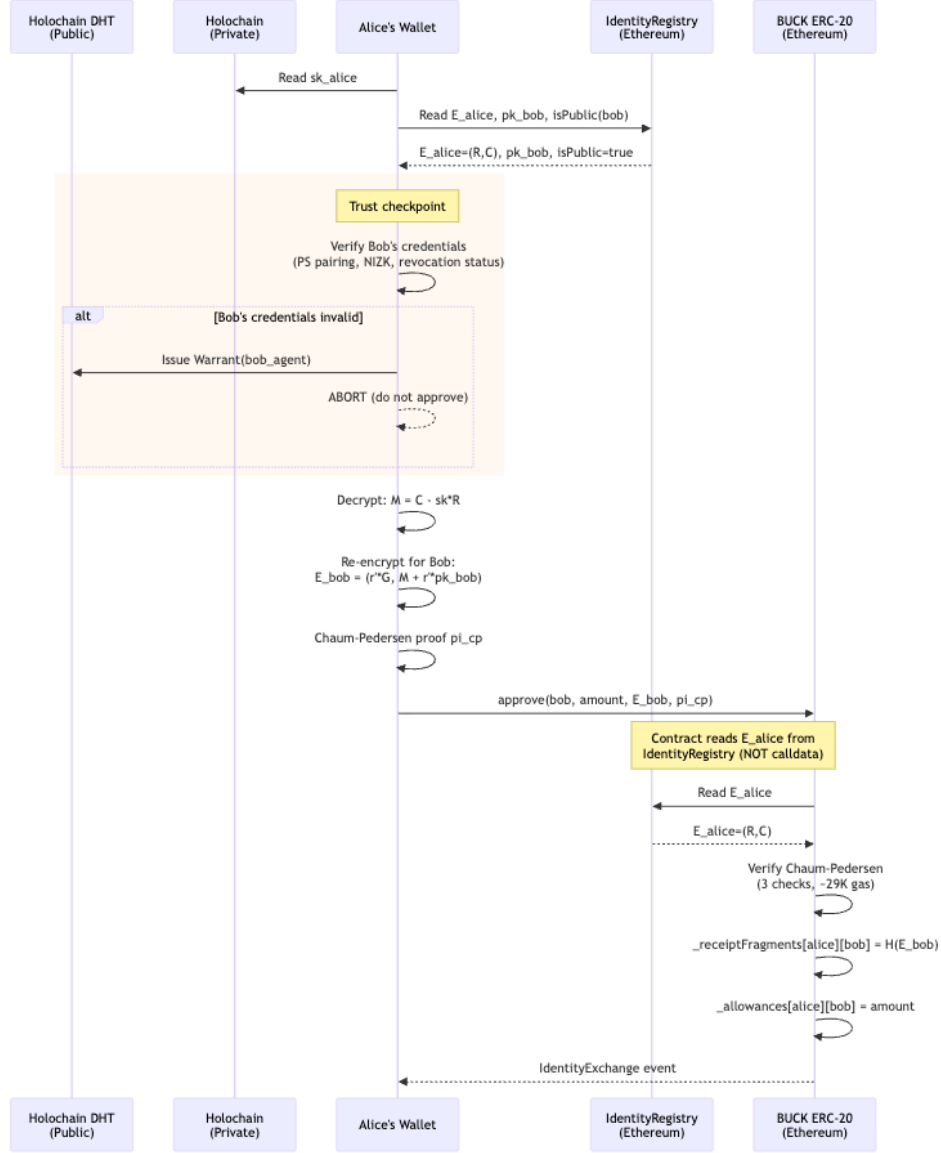
Alice’s wallet reads her Private entries, derives a credential entirely offline, then submits it to the IdentityRegistry. The contract verifies the NIZK proof (~235K gas) and stores the public artifacts. The secret key and plaintext identity *never* touch Ethereum.



1.3 Approve: Re-Encryption with Trust Verification

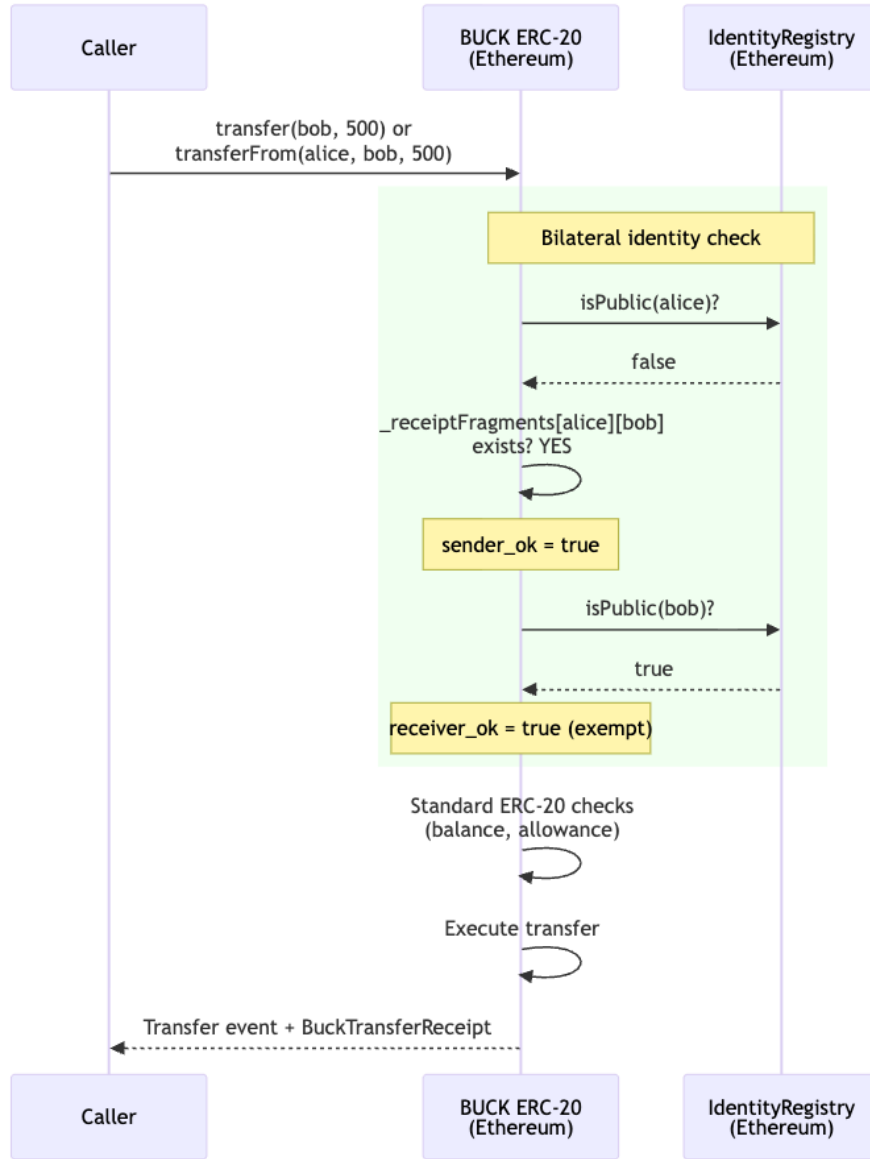
When Alice wants to transact with Bob, her wallet first *verifies Bob's credentials* before re-encrypting. This is the critical trust checkpoint: Alice's wallet reads Bob's on-chain data, checks his PS signature (via the pairing equation) and public key, and only then produces the re-encrypted ciphertext.

If anything is wrong – invalid PS signature, mismatched NIZK proof, revoked credential – Alice simply *does not call approve*. No transaction, no risk. If Alice's wallet detects provable malfeasance (e.g., a forged PS signature from a revoked issuer), it can issue a Holochain **Warrant** against Bob's agent. Multiple independent wallets detecting the same fraud produce consensus: the warranted agent becomes untrusted on the DHT.



1.4 Transfer: Everything On-Chain

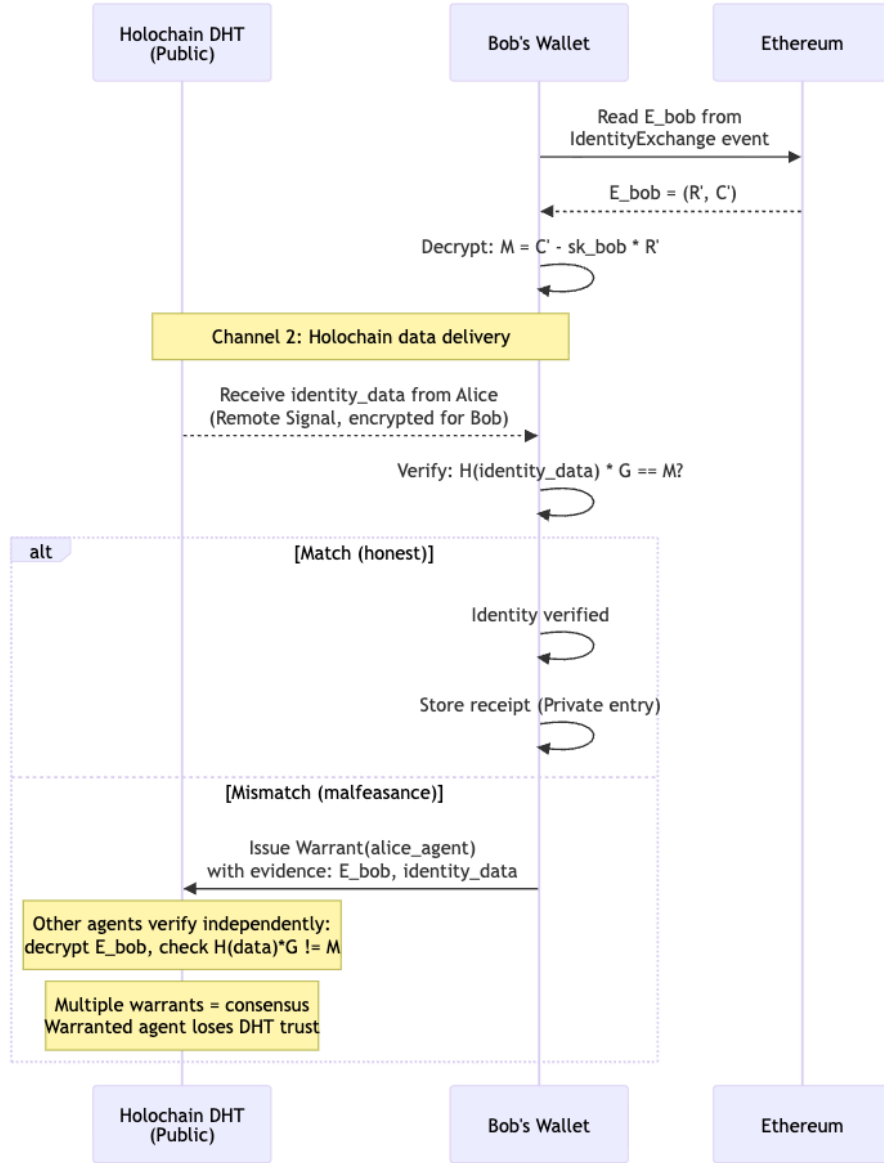
By the time `transfer` or `transferFrom` is called, all proofs have already been verified. The EVM code only checks boolean flags – no cryptography runs during the transfer itself. Public accounts (`isPublic=true`) are exempt from needing a counterparty `approve`.



1.5 Two-Channel Verification and Malfeasance Detection

After the transfer, Bob verifies Alice's identity using two independent channels. Channel 1 (Ethereum) provides the binding proof; Channel 2 (Holochain) provides the content. Neither channel alone is sufficient – this is what makes the design tamper-evident.

If Bob detects that the off-chain `identity_data` doesn't match the on-chain point M , this is *provable* malfeasance: Bob holds the ciphertext (on-chain, timestamped) and the mismatched data (signed by Alice's Holochain agent). Bob issues a Warrant; other agents independently verify the evidence. Since Holochain agents sign their source chain entries, Alice cannot deny having sent the bad data.



1.6 The Four Transfer Modes

The amount of pre-work depends on the visibility of each party:

Sender	Receiver	Who Calls approve	Holochain Work	On-Chain Gas
Private	Private	Both (Alice→Bob, Bob→Alice)	Both exchange via DHT	2 x ~29K
Private	Public	Alice only	Alice sends via DHT	1 x ~29K
Public	Private	Bob only	Bob sends via DHT	1 x ~29K
Public	Public	Neither	None	0 (standard ERC-20)

In every case, the `transfer` / `transferFrom` call itself does *no* cryptography – it only reads the boolean results of prior `approve` calls and `isPublic` flags. All proof verification happens during `approve`.

1.7 The AMM Pool Scenario: Retroactive Identity Recovery

Consider Bob’s AMM liquidity pool – a public BUCK account that processes swaps automatically. Over its lifetime, a million private accounts call `approve(pool, amount, E_pool, proof)` and then swap via `transferFrom`. Bob’s pool contract never examines any counterparty’s identity at the time of the swap. It doesn’t need to: the EVM already verified every proof.

Eighteen months later, a regulator subpoenas Bob for Alice’s identity – account #500,000 – in connection with a fraud investigation. How does Bob comply, and how does the on-chain record prove his pool wasn’t complicit?

1.7.1 What the Chain Already Guarantees

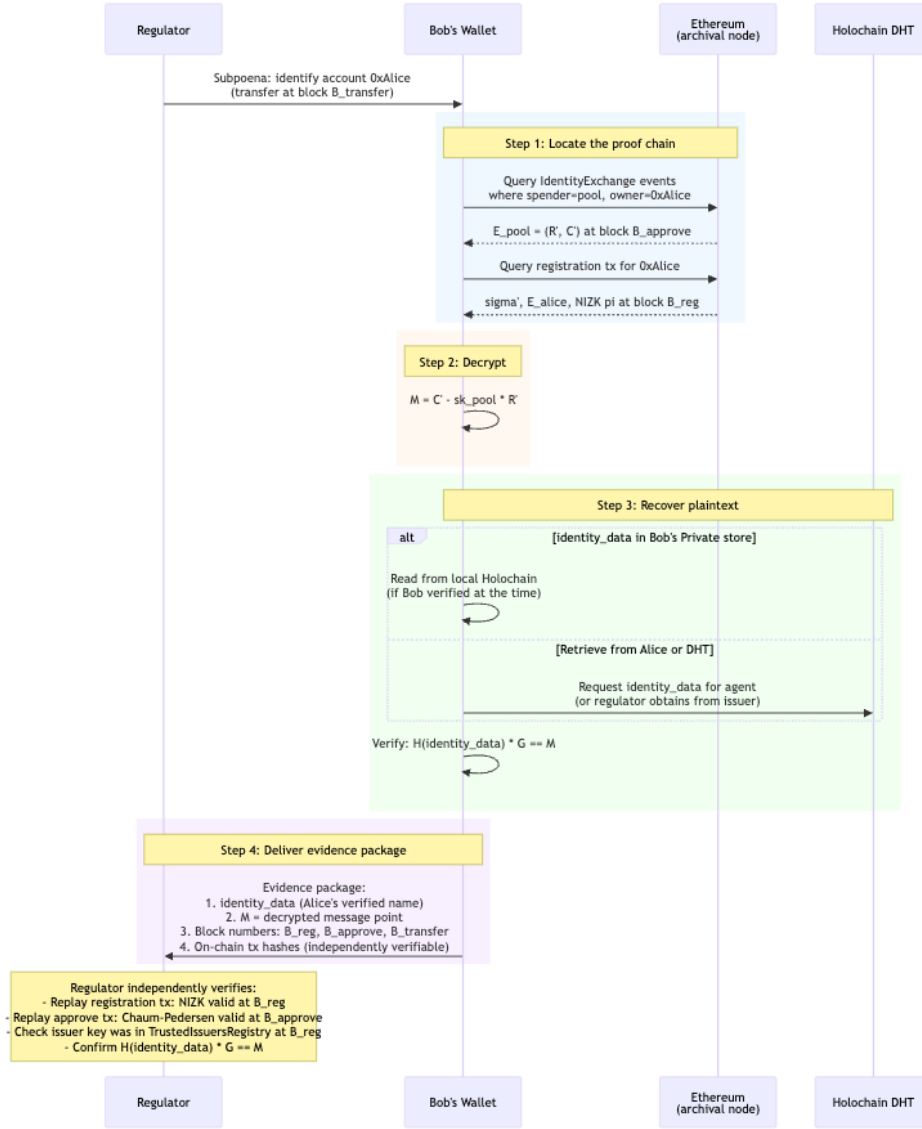
Every `approve` that succeeded is an *immutable proof receipt*. The EVM verified the Chaum-Pedersen proof at that block height; the IdentityRegistry verified Alice’s NIZK at registration time. These facts are encoded in the transaction receipts themselves – the transactions *could not have succeeded* without valid proofs.

The chain of cryptographic commitments is:

1. **Issuance:** A trusted issuer (in TrustedIssuersRegistry) signed Alice’s identity scalar m with a PS signature σ . The issuer’s public key was on-chain and valid at that epoch.
2. **Registration:** The IdentityRegistry verified Alice’s NIZK proof – binding σ' (rerandomized PS signature) to E_{alice} (ElGamal ciphertext). The NIZK proves the *same* m appears in both. The transaction succeeded at block B_{reg} ; the NIZK was valid at that block.
3. **Approve:** The BUCK contract verified Alice’s Chaum-Pedersen proof – proving E_{pool} encrypts the *same* message point M as her registered E_{alice} . The contract read E_{alice} from the IdentityRegistry (not from calldata), so Alice couldn’t substitute a different ciphertext. The transaction succeeded at block B_{approve} .
4. **Transfer:** The bilateral identity check passed. The pool is public (`isPublic(pool) = true`), Alice had a valid `_receiptFragments[alice][pool]`. The swap executed at block B_{transfer} .

None of these steps required Bob to be online, to run a Holochain node, or to examine Alice’s identity. The EVM did it for him.

1.7.2 Bob Responds to the Subpoena



1.7.3 Why Bob Is Protected

Bob's defense is the proof chain itself. He doesn't need to argue that he checked Alice's identity – the *contract* checked it, and the Ethereum transaction log is the evidence:

- **Registration tx succeeded** $\Rightarrow$ NIZK was valid $\Rightarrow$ a trusted issuer certified m at that epoch.
- **Approve tx succeeded** $\Rightarrow$ Chaum-Pedersen was valid $\Rightarrow E_{\text{pool}}$ encrypts the same M as E_{alice} .

- **Bob decrypts** E_{pool} to M ; $M = H(\text{identity_data}) \cdot G$.

The regulator can independently replay these transactions against an archival node. No trust in Bob is required – the math is self-certifying.

Even in adversarial scenarios, Bob is covered:

Scenario	Why Bob is protected
Alice used a stolen identity	The <i>issuer</i> certified it; Bob’s contract verified the issuer’s PS signature was valid at that epoch. Liability falls on the issuer’s KYC process.
Alice’s issuer key was later revoked	The registration tx is timestamped. The issuer key was in TrustedIssuersRegistry at block B_{reg} .
Alice’s identity epoch expired	Epoch expiry is enforced at registration time. If the epoch was valid then, the proof stands.
Alice’s Holochain agent is offline	Bob needs only E_{pool} (on-chain) and sk_pool (his own key). The plaintext can be obtained from the issuer if Alice is uncooperative.
Alice denies the transaction	The approve tx is signed by Alice’s Ethereum account. Non-repudiation is inherent in the blockchain.

1.7.4 Data Availability Over Time

The critical question: will Bob still have what he needs 18 months later?

Data	Availability	Notes
E_{pool} (R’,C’)	Permanent (Ethereum event log)	IdentityExchange event
E_{alice}	Permanent (IdentityRegistry storage)	Until account is closed
σ' , NIZK π	Permanent (registration tx calldata)	Archival node required
Chaum-Pedersen proof	Permanent (approve tx calldata)	Archival node required
sk_pool (Bob’s secret key)	Bob’s custody	Standard key management
identity_data (plaintext)	Holochain DHT (if agent is online) or Bob’s Private store (if he cached)	May require issuer cooperation as fallback
Issuer public key history	TrustedIssuersRegistry (on-chain)	Includes revocation timestamps

Everything except the plaintext **identity_data** is permanently on-chain. The plaintext is the one piece that lives off-chain – but Bob only needs it to produce a *human-readable name* for the regulator. The cryptographic proof chain stands without it: M is recoverable from E_{pool} alone, and M is what the issuer certified.

If Alice’s Holochain agent is long gone, the regulator can obtain **identity_data** from the issuer (who has it from the original KYC ceremony) and verify $H(\text{identity_data}) \cdot G = M$ independently. The issuer doesn’t need Bob’s cooperation; the regulator doesn’t need Alice’s.

1.7.5 Why M Is the Same Across All Accounts – And Why That’s Correct

A natural concern: if Alice derives ten accounts from the Identity Fountain, and each counterparty who decrypts their re-encrypted ciphertext recovers the same $M = m \cdot G$, doesn’t that break unlinkability?

No. The Identity Fountain’s guarantee is about *on-chain unlinkability* – what a passive observer (Eve) can learn from the public blockchain data alone, without any secret keys. That guarantee is airtight:

- σ'_1 and σ'_2 are *statistically* unlinkable (information-theoretic, not merely computational)
- E_1 and E_2 are IND-CPA secure under independent keys
- pk_1 and pk_2 are independent random points

No algorithm, no amount of computation, can correlate Alice’s accounts from on-chain data.

The identity scalar $m = H(\text{identity_data})$ is *intentionally invariant* because it *is* Alice’s identity. If each account used a different m , the two-channel verification would break: Bob couldn’t verify that the `identity_data` he received matches what the issuer certified. The Identity Fountain protects the *credential wrapping* (σ', pk, E) , not the identity itself.

Think of it as giving Alice a new disguise for each account – different keys, different signature, different ciphertext. The disguises fool passive surveillance (Eve). They don’t fool someone Alice voluntarily shows her face to (Bob, during **approve**).

Observer	Can link accounts?	Why?
Eve (passive, on-chain)	NO	All public artifacts are independent random values
Bob (authorized)	Learns one account	By design – the point of approve
Bob + Carol (colluding)	Yes, via M	But both already know Alice’s name
Issuer (on-chain only)	NO	Can’t extract M without a secret key
Issuer (given M)	YES	By design – regulatory compliance path

The privacy boundary is clear: *before* **approve**, Alice’s accounts are unlinkable to everyone. *After* **approve**, the specific counterparty learns Alice’s identity for that account – and only that counterparty (plus anyone with whom they share M , but they already shared Alice’s name via the `identity_data` exchange).

1. Alice Tries to Hide by Not Publishing `identity_data`

Alice might try to game the system: complete the on-chain **approve** (re-encryption + Chaum-Pedersen proof), satisfying the ERC-20’s bilateral identity check, but never publish her `identity_data` to the Holochain DHT. She hopes that if a subpoena arrives, Bob won’t be able to produce a human-readable name.

This doesn’t work. Bob still holds the cryptographic thread:

- (a) Bob decrypts E_{pool} to recover $M = m \cdot G$. This requires only Bob’s secret key and the on-chain ciphertext – Alice’s cooperation is unnecessary.
- (b) The regulator takes M to the issuer. The issuer computes $m_i \cdot G$ for each identity they have issued and finds the match. The issuer has Alice’s **identity_data** from the original KYC ceremony.
- (c) The issuer confirms: this M belongs to Alice Johnson.

Alice’s attempt to hide by not publishing is futile: the on-chain E_{pool} and the issuer’s records are sufficient. Alice cannot retroactively alter either.

2. Bob’s Wallet Detects Non-Publication

Bob’s wallet (Holochain hApp) can proactively monitor for counterparties who complete the on-chain **approve** but fail to publish the required DHT entries. The expected protocol is:

- (a) Alice calls **approve(pool, amount, E_pool, proof)** on Ethereum.
- (b) Alice’s wallet publishes a DHT entry containing her **identity_data**, encrypted for Bob’s Holochain agent, and keyed by the **IdentityExchange** event hash.
- (c) Bob’s wallet watches for the **IdentityExchange** event and looks up the corresponding DHT entry within a grace period.

If step 2 never happens – the on-chain **approve** succeeded but no DHT entry appears – Bob’s wallet can:

- Flag the account as non-compliant (Alice approved on-chain but didn’t complete the Holochain identity exchange).
- Issue a Holochain **Warrant** against Alice’s agent, citing the on-chain tx hash as evidence of protocol violation.
- Report the account to pool operators or compliance systems.
- Optionally, Bob’s contract could revoke the approval (`reset_receiptFragments[alice][pool]`), blocking future swaps until Alice completes the full protocol.

The warrant is independently verifiable: any Holochain agent can check that the **IdentityExchange** event exists on Ethereum but the corresponding DHT entry does not. Multiple independent warrants produce DHT-level consensus that Alice’s agent is non-compliant.

Even if Alice never publishes, Bob’s regulatory position is unchanged. He can still decrypt M from the on-chain ciphertext and hand it to the regulator, who obtains **identity_data** from the issuer. Alice’s non-publication delays the human-readable identification by one step (issuer lookup) but cannot prevent it.

2 Setup: The alt_bn128 Curve and Helpers

All operations use BN254, the curve behind Ethereum's `ecAdd` (0x06), `ecMul` (0x07), and `ecPairing` (0x08) precompiles. The `py_ecc.bn128` library provides exactly these primitives. Helper functions for random scalar generation, `keccak256` hashing, and point formatting are defined here; all subsequent code blocks share a single `:session` `abid` so that variables persist across steps.

```
import secrets
from py_ecc.bn128 import bn128_curve as bc, bn128_pairing as bp
from web3 import Web3

G1      = bc.G1          # Generator of G_1
G2      = bc.G2          # Generator of G_2
Z1      = bc.Z1          # Point at infinity (G_1 identity)
ORDER   = bc.curve_order # Curve group order (prime)
mul      = bc.multiply   # Scalar multiplication in G_1 or G_2
add      = bc.add        # Point addition
neg      = bc.neg        # Point negation
eq       = bc.eq         # Point equality
pairing  = bp.pairing    # Optimal Ate pairing: G_2 x G_1 -> G_T

def rand():
    """Random non-zero scalar mod ORDER."""
    return secrets.randbelow(ORDER - 1) + 1

def keccak(*args):
    """keccak256 over concatenated big-endian 32-byte words, reduced mod ORDER."""
    data = b''.join(v.to_bytes(32, 'big') if isinstance(v, int)
                    else str(v).encode() for v in args)
    h = Web3.solidity_keccak(['bytes'], [data])
    return int.from_bytes(h, 'big') % ORDER

def pt(P, label=""):
    """Format a G_1 point for display."""
    if bc.is_inf(P):
        return f"{label}0 (infinity)"
    return f"{label}({int(P[0]) % 10**6:>6d}..., {int(P[1]) % 10**6:>6d}...)"

print(f"BN254 curve order: {ORDER}")
print(f"G_1 generator:      {pt(G1)}")
print(f"G_2 generator:      (pair of FQ2 elements)")

BN254 curve order: 21888242871839275222246405745257275088548364400416034343698204186575808495617
G_1 generator:      ( 1..., 2...)
G_2 generator:      (pair of FQ2 elements)
```

3 Issuer Key Generation

The identity issuer (e.g., ATB Financial acting as Alberta's KYC authority) generates a Pointcheval-Sanders key pair: secret key (x, y) (two scalars), public key (X, Y) (two points in G_2 , published on-chain in the `TrustedIssuersRegistry`). This key pair is the root of trust for all identities the issuer certifies; every credential in the system traces back to a PS signature under this key.

```

# Issuer's PS secret key
isk_x = rand()
isk_y = rand()

# Issuer's PS public key (in G_2, published on-chain in TrustedIssuersRegistry)
IPK_X = mul(G2, isk_x)
IPK_Y = mul(G2, isk_y)

print("Issuer PS key pair generated.")
print(f"  sk = (x, y)                -- two secret scalars")
print(f"  PK = (X, Y) in G_2          -- published in TrustedIssuersRegistry")

Issuer PS key pair generated.
  sk = (x, y)                -- two secret scalars
  PK = (X, Y) in G_2          -- published in TrustedIssuersRegistry

```

4 KYC Ceremony – Issuer Signs Alice’s Identity

Alice presents her identity documents to the issuer, who verifies them, constructs a canonical `identity_data` JSON structure (sorted keys, no whitespace), and computes the identity scalar $m = \text{keccak256}(\text{identity_data}) \bmod \text{ORDER}$. The issuer then produces a PS signature $\sigma = (h, (x + m \cdot y) \cdot h)$ binding m to the issuer’s secret key. The bilinear pairing equation verifies the signature, and a counter-example confirms that any change to `identity_data` (producing a different m) causes verification to fail.

4.1 Alice’s Plaintext Identity

```

import json

# Alice's verified identity (canonical encoding -- field order matters)
alice_identity_data = json.dumps({
    "given_name": "Alice",
    "family_name": "Johnson",
    "jurisdiction": "Alberta, Canada",
    "id_type": "Alberta Identity Card",
    "id_number": "AIC-2026-4839201",
    "date_of_birth": "1992-03-15",
    "issuer_id": "atb-financial-ca",
    "issued_at": "2026-01-20T14:30:00Z",
    "epoch": 42
}, sort_keys=True, separators=(',', ':'))

print("Alice's identity_data (canonical JSON):")
print(f"  {alice_identity_data[:72]}...")

Alice's identity_data (canonical JSON):
  {"date_of_birth":"1992-03-15","epoch":42,"family_name":"Johnson","given_...

```

4.2 Compute Identity Scalar and PS Signature

The PS signature is $\sigma = (h, (x + m \cdot y) \cdot h)$ where h is a random point in G_1 .

```

# Identity scalar: m = keccak256(identity_data) mod ORDER
m_alice = keccak(alice_identity_data)

# PS signature: pick random h, compute sigma = (h, (x + m*y) * h)
t_h      = rand()
h        = mul(G1, t_h)                # random G_1 point
sigma_2 = mul(h, (isk_x + m_alice * isk_y) % ORDER)
sigma    = (h, sigma_2)

print(f"Identity scalar m = H(identity_data):")
print(f"  m = {m_alice}")
print(f"PS signature sigma:")
print(f"  sigma_1 = {pt(sigma[0])}")
print(f"  sigma_2 = {pt(sigma[1])}")

Identity scalar m = H(identity_data):
  m = 6446313962173165014188361468977035479170665192834402473818305319846425860127
PS signature sigma:
  sigma_1 = (355897..., 626966...)
  sigma_2 = (261916..., 860983...)

```

4.3 Verify the PS Signature (Issuer Self-Check)

The PS verification equation uses a bilinear pairing:

$$e(\sigma_1, X + m \cdot Y) = e(\sigma_2, g_2)$$

What this proves: The issuer's signature is mathematically valid – it binds the identity scalar m to the issuer's secret key. If the signature passes, Alice can be sure the issuer actually certified her identity (no one else knows (x, y)).

```

# Verify: e(sigma_1, X + m*Y) == e(sigma_2, g_2)
lhs = pairing(add(IPK_X, mul(IPK_Y, m_alice)), sigma[0])
rhs = pairing(G2, sigma[1])
assert lhs == rhs, "PS signature verification FAILED"
print("PS signature verified: e(sigma_1, X + m*Y) == e(sigma_2, g_2)")

# The issuer gives Alice: (m, sigma, identity_data)
# Alice stores all three in her Holochain Private entry
print(f"\nIssuer gives Alice:")
print(f"  m          = {m_alice}")
print(f"  sigma       = (sigma_1, sigma_2)  -- PS signature")
print(f"  identity_data = '{alice_identity_data[:40]}...'")

PS signature verified: e(sigma_1, X + m*Y) == e(sigma_2, g_2)

Issuer gives Alice:
  m          = 6446313962173165014188361468977035479170665192834402473818305319846425860127
  sigma       = (sigma_1, sigma_2)  -- PS signature
  identity_data = '{"date_of_birth":"1992-03-15","epoch":42...}'

```

How to read the output: "PS signature verified" means both sides of the pairing equation produce the same element in G_T . If m were wrong (e.g., tampered identity data), the two pairings would produce *different* G_T elements and the assertion would fail. A counter-example follows.

4.3.1 Counter-example: Wrong Identity Scalar

If Eve tries to forge a signature using a different m , the pairing equation fails. This is what a rejected verification looks like:

```
# Tampered m -- what if someone changes one field in identity_data?
m_fake = keccak("tampered_identity_data")
lhs_bad = pairing(add(IPK_X, mul(IPK_Y, m_fake)), sigma[0])
rhs_good = pairing(G2, sigma[1])
ok = (lhs_bad == rhs_good)
print(f"PS verify with wrong m: {ok}")
print(f" (The two G_T elements differ -- the pairing detects the mismatch.)")
print(f" Any change to identity_data produces a different m, which")
print(f" makes the pairing equation fail. The signature is unforgeable.")
```

```
PS verify with wrong m: False
(The two G_T elements differ -- the pairing detects the mismatch.)
Any change to identity_data produces a different m, which
makes the pairing equation fail. The signature is unforgeable.
```

5 The Identity Fountain – Alice Derives a Fresh Credential

Alice wants to create a new Ethereum account. Her Holochain wallet runs the Identity Fountain entirely offline, producing four artifacts: a rerandomized PS signature σ' (statistically unlinkable to the original), a fresh key pair (sk, pk) , an ElGamal encryption of m under pk , and a NIZK sigma-protocol proof binding the PS signature to the ciphertext. Each step is demonstrated below: rerandomization preserves signature validity while destroying correlation, and ElGamal decryption recovers the correct message point. A counter-example confirms that the wrong secret key yields an unrelated random point.

5.1 Rerandomize the PS Signature

Alice picks a random scalar t and computes $\sigma' = (t \cdot \sigma_1, t \cdot \sigma_2)$. This is a valid PS signature on the same m – but *statistically unlinkable* to the original σ .

What this proves: Rerandomization produces a fresh, valid PS signature that cannot be correlated with the original – even by the issuer.

```
# Rerandomize: sigma' = (t * sigma_1, t * sigma_2)
t_rerand = rand()
sigma_p_1 = mul(sigma[0], t_rerand)
sigma_p_2 = mul(sigma[1], t_rerand)
sigma_p = (sigma_p_1, sigma_p_2)

# Verify sigma' is still a valid PS signature on m
lhs = pairing(add(IPK_X, mul(IPK_Y, m_alice)), sigma_p[0])
rhs = pairing(G2, sigma_p[1])
assert lhs == rhs, "Rerandomized PS signature FAILED"
assert not bc.is_inf(sigma_p[0]), "sigma'_1 must not be point at infinity"
print("Rerandomized PS signature verified.")
print(f" sigma'_1 = {pt(sigma_p[0])}")
print(f" sigma'_2 = {pt(sigma_p[1])}")
print(f"\nOriginal and rerandomized signatures are unlinkable:")
print(f" sigma_1 = {pt(sigma[0])}")
print(f" sigma'_1 = {pt(sigma_p[0])}")
print(f" (No algorithm can correlate them.)")
```

```

Rerandomized PS signature verified.
sigma'_1 = (277377..., 422277...)
sigma'_2 = (491255..., 15354...)

Original and rerandomized signatures are unlinkable:
sigma_1 = (355897..., 626966...)
sigma'_1 = (277377..., 422277...)
(No algorithm can correlate them.)

```

How to read the output: Two things to check: (1) "Rerandomized PS signature verified" confirms σ' still satisfies the pairing equation. (2) Compare the coordinate fragments of `sigma_1` vs. `sigma'_1` – they share no visible pattern. This isn't coincidence: the rerandomization is *statistical* (information-theoretic), not merely computational.

5.2 Generate Fresh Identity Key Pair and ElGamal Encryption

Alice generates a new key pair on `alt_bn128` and encrypts her identity scalar m under her own public key using ElGamal:

$$E_{\text{alice}} = (r \cdot G, m \cdot G + r \cdot pk_{\text{alice}})$$

What this proves: Alice can encrypt her identity under her own public key and recover it. ElGamal decryption yields the *point* $M = m \cdot G$, not the scalar m (solving the discrete log is infeasible).

```

# Fresh identity key pair for this account
sk_alice = rand()
pk_alice = mul(G1, sk_alice)

# ElGamal encryption of m under pk_alice
r_enc = rand()
R_alice = mul(G1, r_enc)
M_alice = mul(G1, m_alice)
C_alice = add(M_alice, mul(pk_alice, r_enc))
E_alice = (R_alice, C_alice)

# R = r * G
# M = m * G (message point)
# C = m*G + r*pk

print(f"Fresh key pair for this Ethereum account:")
print(f"  sk_alice = {sk_alice}")
print(f"  pk_alice = {pt(pk_alice)}")
print(f"\nElGamal ciphertext E_alice = (R, C):")
print(f"  R = r*G      = {pt(R_alice)}")
print(f"  C = m*G+r*pk = {pt(C_alice)}")
print(f"\nMessage point M = m*G:")
print(f"  M = {pt(M_alice)}")

# Verify Alice can decrypt her own ciphertext
M_check = add(C_alice, neg(mul(R_alice, sk_alice))) # C - sk*R = M
assert eq(M_check, M_alice), "ElGamal self-decryption FAILED"
print(f"\nAlice decrypts: C - sk*R = {pt(M_check)} (matches M)")

Fresh key pair for this Ethereum account:
sk_alice = 2527918879824434158347107444152090048840608521824691372403084309625630276238
pk_alice = (561696..., 359461...)

ElGamal ciphertext E_alice = (R, C):
R = r*G      = (778258..., 705687...)

```

```

C = m*G+r*pk = (514843..., 322599...)

Message point M = m*G:
M = (739132..., 937433...)

Alice decrypts: C - sk*R = (739132..., 937433...) (matches M)

```

How to read the output: The last two lines show the decrypted point $C - sk \cdot R$ and the original M . Their coordinate fragments must match – this confirms the ciphertext is well-formed. If Alice used the wrong secret key, the decrypted point would be a random, unrelated point (shown in the counter-example below).

5.2.1 Counter-example: Wrong Secret Key

An attacker who doesn't know `sk_alice` cannot decrypt to the correct M . The result is an unrelated random point:

```

sk_eve = rand() # Eve's key, not Alice's
M_wrong = add(C_alice, neg(mul(R_alice, sk_eve)))
print(f"Eve decrypts with wrong key:")
print(f"  M_wrong = {pt(M_wrong)}")
print(f"  M_real  = {pt(M_alice)}")
print(f"  Match: {eq(M_wrong, M_alice)}")
print(f"  (Completely different point -- no information about m is leaked.)")

```

```

Eve decrypts with wrong key:
M_wrong = (150076..., 124946...)
M_real  = (739132..., 937433...)
Match: False
(Completely different point -- no information about m is leaked.)

```

5.3 NIZK Proof: Binding PS Signature to ElGamal Ciphertext

Alice must prove (without revealing m or r) that:

- (a) σ' is a valid PS signature on m
- (b) E_{alice} encrypts the same m

This is a Schnorr-family sigma protocol. The PS relation is *linear* in m , so no Groth-Sahai proofs or SNARKs are needed.

```

# === PROVER (Alice, offline) ===
# Commit: pick random blinding scalars
m_tilde = rand() # blinding for m
r_tilde = rand() # blinding for r

# Commitment for PS equation (a): A_ps = m_tilde * sigma'_1
# (This will be checked against the pairing equation)
A_ps = mul(sigma_p[0], m_tilde)

# Commitments for ElGamal equations (b) and (c):
T_C = add(mul(G1, m_tilde), mul(pk_alice, r_tilde)) # m~*G + r~*pk
T_R = mul(G1, r_tilde)                             # r~*G

```

```

# Fiat-Shamir challenge: hash all public inputs + commitments
# (In Ethereum: includes registrant_address for domain separation)
e = keccak(
    sigma_p[0][0], sigma_p[0][1], # sigma'_1
    sigma_p[1][0], sigma_p[1][1], # sigma'_2
    R_alice[0], R_alice[1],       # E_alice.R
    C_alice[0], C_alice[1],       # E_alice.C
    pk_alice[0], pk_alice[1],     # pk_alice
    A_ps[0], A_ps[1],             # PS commitment
    T_C[0], T_C[1],               # ElGamal C commitment
    T_R[0], T_R[1],               # ElGamal R commitment
)

# Responses
s_m = (m_tilde + e * m_alice) % ORDER
s_r = (r_tilde + e * r_enc) % ORDER

pi = (e, s_m, s_r, A_ps, T_C, T_R)
print("NIZK proof pi = (e, s_m, s_r, A_ps, T_C, T_R)")
print(f" challenge e = {e}")
print(f" response s_m = {s_m}")
print(f" response s_r = {s_r}")

NIZK proof pi = (e, s_m, s_r, A_ps, T_C, T_R)
challenge e = 2010956954307822119432409248316033680714438509237249716458726920919524357382
response s_m = 9415020179305125084988302176808802560575060940531678030578914517956884891628
response s_r = 753420188019358874828155111641290499339048010712326623128412636908358858056

```

6 Registration – On-Chain Verification

Alice submits $(pk, E_{\text{alice}}, \sigma', \pi)$ to the IdentityRegistry contract. The contract runs five verification checks using the `ecPairing`, `ecMul`, and `ecAdd` precompiles ($\sim 235\text{K}$ gas total): two ElGamal consistency checks, one PS pairing product check, a Fiat-Shamir challenge reconstruction, and a non-triviality guard. All five must pass for `isVerified[alice]` to be set. The full verification runs below, followed by a counter-example where Alice tries to register an ElGamal ciphertext containing a different m than her PS signature – the ElGamal consistency check catches the mismatch.

6.1 Verify the NIZK Proof (What the Contract Computes)

What this proves: The same m appears in both the PS signature and the ElGamal ciphertext, without revealing m or r . Five checks run:

- (b) ElGamal C consistency: the committed m is inside C
- (c) ElGamal R consistency: the committed r is inside R
- (a) PS pairing product: the same m satisfies the PS equation
- (d) Fiat-Shamir: the challenge was honestly derived (not chosen after the fact)
- (e) Non-triviality: $\sigma'_1 \neq \mathcal{O}$ (prevents trivial forgery)

```

# === VERIFIER (IdentityRegistry contract, on-chain) ===
# Unpack proof
e_v, s_m_v, s_r_v, A_ps_v, T_C_v, T_R_v = pi

# --- Check (b): ElGamal consistency ---
# Verify: s_m * G + s_r * pk_alice == e * C + T_C
lhs_C = add(mul(G1, s_m_v), mul(pk_alice, s_r_v))
rhs_C = add(mul(C_alice, e_v), T_C_v)
assert eq(lhs_C, rhs_C), "ElGamal C check FAILED"
print("(b) ElGamal C check passed: s_m * G + s_r * pk == e * C + T_C")

# Verify: s_r * G == e * R + T_R
lhs_R = mul(G1, s_r_v)
rhs_R = add(mul(R_alice, e_v), T_R_v)
assert eq(lhs_R, rhs_R), "ElGamal R check FAILED"
print("(c) ElGamal R check passed: s_r * G == e * R + T_R")

# --- Check (a): PS signature via pairing ---
# Reconstruct PS commitment: s_m * sigma'_1 should equal A_ps + e * sigma'_2
# relative to the pairing equation.
# Verify pairing product:
#   e(s_m * sigma'_1, Y) * e(-A_ps, Y) * e(e * sigma'_1, X)
#   * e(-e * sigma'_2, g_2) == 1
# Which simplifies to:
#   e(s_m * sigma'_1 - A_ps, Y) * e(e * sigma'_1, X) * e(-e * sigma'_2, g_2) == 1
# This is equivalent to checking:
#   e(A_ps, Y) * e(e * sigma'_2, g_2) == e(s_m * sigma'_1, Y) * e(e * sigma'_1, X)

P1 = mul(sigma_p[0], s_m_v)          # s_m * sigma'_1
P2 = add(A_ps_v, mul(sigma_p[1], e_v)) # A_ps + e * sigma'_2
P3 = mul(sigma_p[0], e_v)           # e * sigma'_1

# Pairing check: e(P1, Y) == e(P2, Y) * e(P3, X) / e(P3, X)
# More directly: verify the original PS equation holds through the proof
# e(s_m * sigma'_1 - e * sigma'_2, Y) == e(A_ps + e * sigma'_1, X) ... etc.
#
# Simplified: check that the reconstructed pairing equation holds:
# e(sigma'_1, X + m * Y) = e(sigma'_2, g_2) via the proof responses
#
# The verifier checks: e(s_m * s'_1, Y) * e(e * s'_1, X) == e(A_ps, Y) * e(e * s'_2, g_2)
lhs_pair = pairing(IPK_Y, P1) * pairing(IPK_X, P3)
rhs_pair = pairing(IPK_Y, P2) * pairing(G2, mul(sigma_p[1], e_v))
# Actually let's use the direct verification approach:
# We know s_m = m~ + e * m. The commitment A_ps = m~ * sigma'_1.
# So s_m * sigma'_1 = m~ * sigma'_1 + e * m * sigma'_1 = A_ps + e * m * sigma'_1
# The verifier doesn't know m, but can check via pairing:
# e(s_m * sigma'_1, Y) * e(e * sigma'_1, X) * e(-e * sigma'_2 - A_ps, g_2_placeholder)
# This gets complex in code. Let's use the direct pairing product check.

# Direct approach: verify e(sigma'_1, X + m * Y) = e(sigma'_2, g_2)
# by extracting m from the sigma protocol responses:
# The key insight: the SAME s_m that satisfies (b) must also satisfy (a).
# If checks (b) and (c) pass, we know s_m = m~ + e * m for the m in E_alice.
# Now check (a): does this same m satisfy the PS equation?
#
# Reconstruct: m * sigma'_1 = (s_m * sigma'_1 - A_ps) / e
# But we avoid division. Instead verify the pairing product equation:
#   e(s_m * sigma'_1 - A_ps, Y) == e(e * sigma'_2, g_2) * e(-e * sigma'_1, X)
#   (rearranging e(sigma'_1, X + m * Y) = e(sigma'_2, g_2) with substitution)

LHS_point = add(mul(sigma_p[0], s_m_v), neg(A_ps_v)) # s_m * s'_1 - A_ps = e * m * s'_1
RHS_neg    = add(mul(sigma_p[1], e_v), neg(mul(sigma_p[0], e_v))) # e * s'_2 - e * s'_1

```

```

# not quite right -- use the clean verification instead

# Clean pairing product check (what ecPairing precompile computes):
# e(s_m*sigma'_1, Y) * e(-A_ps, Y) * e(e*s'_1, X) * e(-e*s'_2, g_2) == 1
check = (
    pairing(IPK_Y, mul(sigma_p[0], s_m_v))      # e(s_m*s'_1, Y)
    * pairing(IPK_Y, neg(A_ps_v))              # e(-A_ps, Y)
    * pairing(IPK_X, mul(sigma_p[0], e_v))      # e(e*s'_1, X)
    * pairing(G2, neg(mul(sigma_p[1], e_v)))    # e(-e*s'_2, g_2)
)
assert check == bp.FQ12.one(), "PS pairing check FAILED"
print("(a) PS pairing check passed:")
print("    e(s_m*s'_1, Y) * e(-A_ps, Y) * e(e*s'_1, X) * e(-e*s'_2, g_2) == 1")

# --- Fiat-Shamir challenge reconstruction ---
e_reconstructed = keccak(
    sigma_p[0][0], sigma_p[0][1],
    sigma_p[1][0], sigma_p[1][1],
    R_alice[0], R_alice[1],
    C_alice[0], C_alice[1],
    pk_alice[0], pk_alice[1],
    A_ps_v[0], A_ps_v[1],
    T_C_v[0], T_C_v[1],
    T_R_v[0], T_R_v[1],
)
assert e_reconstructed == e_v, "Fiat-Shamir challenge mismatch"
print("(d) Fiat-Shamir challenge verified (binds all public inputs)")

# --- sigma'_1 != 0 check ---
assert not bc.is_inf(sigma_p[0]), "sigma'_1 is point at infinity (trivial forgery)"
print("(e) sigma'_1 != 0 (non-trivial signature)")

print("\n==> Registration PASSED. isVerified(alice) = true")

(b) ElGamal C check passed: s_m*G + s_r*pk == e*C + T_C
(c) ElGamal R check passed: s_r*G == e*R + T_R
(a) PS pairing check passed:
    e(s_m*s'_1, Y) * e(-A_ps, Y) * e(e*s'_1, X) * e(-e*s'_2, g_2) == 1
(d) Fiat-Shamir challenge verified (binds all public inputs)
(e) sigma'_1 != 0 (non-trivial signature)

==> Registration PASSED. isVerified(alice) = true

```

How to read the output: All five checks must print "passed". If any prints "FAILED", the proof is invalid. A counter-example follows showing what happens when Alice tries to register a ciphertext containing a *different* m than the one in her PS signature.

6.1.1 Counter-example: ElGamal Encrypts a Different m

If Alice (or an attacker) tries to encrypt one m in ElGamal but prove a PS signature on a *different* m , the ElGamal consistency check detects the mismatch:

```

# Forge: encrypt a fake m_fake in ElGamal, but use the real PS sigma
m_fake = rand()
r_fake = rand()
R_fake = mul(G1, r_fake)
C_fake = add(mul(G1, m_fake), mul(pk_alice, r_fake))

```

```

# Try to build a NIZK proof binding sigma' (real m) to E_fake (fake m)
m_t = rand(); r_t = rand()
A_fake = mul(sigma_p[0], m_t)
T_C_f = add(mul(G1, m_t), mul(pk_alice, r_t))
T_R_f = mul(G1, r_t)
e_f = keccak(
    sigma_p[0][0], sigma_p[0][1], sigma_p[1][0], sigma_p[1][1],
    R_fake[0], R_fake[1], C_fake[0], C_fake[1],
    pk_alice[0], pk_alice[1],
    A_fake[0], A_fake[1], T_C_f[0], T_C_f[1], T_R_f[0], T_R_f[1])
s_m_f = (m_t + e_f * m_alice) % ORDER # uses real m for PS...
s_r_f = (r_t + e_f * r_fake) % ORDER # ...but fake r for ElGamal

# Check (b): s_m*G + s_r*pk == e*C_fake + T_C?
lhs = add(mul(G1, s_m_f), mul(pk_alice, s_r_f))
rhs = add(mul(C_fake, e_f), T_C_f)
ok = eq(lhs, rhs)
print(f"Forged proof -- ElGamal C check: {ok}")
print(f" (The check FAILS because s_m encodes m_real but C_fake")
print(f" encrypts m_fake. The NIZK forces both to be identical.)")

```

```

Forged proof -- ElGamal C check: False
(The check FAILS because s_m encodes m_real but C_fake
encrypts m_fake. The NIZK forces both to be identical.)

```

7 Bob Registers as a Public Account

Bob operates a Uniswap-style AMM liquidity pool and registers as a public BUCK account: his `identity_data` is world-readable in the IdentityRegistry, and `isPublic(bob) = true`. This exempts Bob's counterparties from needing Bob to call `approve` – any private account can approve Bob's pool unilaterally, and the bilateral identity check in `transfer` passes because Bob's public status satisfies his side automatically.

```

# Bob's identity (public pool operator)
bob_identity_data = json.dumps({
    "given_name": "Bob",
    "family_name": "Smith",
    "jurisdiction": "Alberta, Canada",
    "id_type": "Corporate Registration",
    "id_number": "AB-CORP-2026-00182",
    "date_of_birth": "1985-07-22",
    "issuer_id": "atb-financial-ca",
    "issued_at": "2026-02-01T09:00:00Z",
    "epoch": 42
}, sort_keys=True, separators=(',', ':'))

m_bob = keccak(bob_identity_data)

# Bob also gets a PS signature (same issuance ceremony)
t_h_bob = rand()
h_bob = mul(G1, t_h_bob)
sigma_bob = (h_bob, mul(h_bob, (isk_x + m_bob * isk_y) % ORDER))

# Bob generates key pair and ElGamal encryption
sk_bob = rand()
pk_bob = mul(G1, sk_bob)
r_bob = rand()

```

```

R_bob = mul(G1, r_bob)
M_bob = mul(G1, m_bob)
C_bob = add(M_bob, mul(pk_bob, r_bob))
E_bob_reg = (R_bob, C_bob)

# Bob registers (same PS+NIZK verification as Alice -- omitted for brevity)
# AND publishes identity_data on-chain (public account)
print("Bob registered as PUBLIC account:")
print(f" pk_bob = {pt(pk_bob)}")
print(f" E_bob = (R, C) -- ElGamal ciphertext on-chain")
print(f" identity_data = '{bob_identity_data[:50]}...'")
print(f" isVerified(bob) = true")
print(f" isPublic(bob) = true")
print(f"\nBob's identity is world-readable. No approve() needed from Bob's side.")

Bob registered as PUBLIC account:
pk_bob = (634253..., 878599...)
E_bob = (R, C) -- ElGamal ciphertext on-chain
identity_data = '{"date_of_birth":"1985-07-22","epoch":42,"family_n...'}
isVerified(bob) = true
isPublic(bob) = true

Bob's identity is world-readable. No approve() needed from Bob's side.

```

8 Alice Approves Bob – Re-Encryption + Chaum-Pedersen Proof

Alice wants to swap BUCK for USDT on Bob's pool. She calls `approve(pool, amount, E_pool, proof)` to re-encrypt her identity under Bob's public key and prove the re-encryption is correct via a Chaum-Pedersen proof (~29K gas). Since Bob is public, only Alice needs to approve; the bilateral identity check in `transfer` passes because `isPublic(bob) = true` satisfies Bob's side automatically. Below: the re-encryption, the three-check Chaum-Pedersen verification, and a counter-example showing that re-encrypting a *different* message point causes Check 2 to fail.

8.1 Re-Encrypt Alice's Identity for Bob

Alice decrypts her own ciphertext to recover $M = m \cdot G$, then re-encrypts under Bob's public key with fresh randomness.

What this proves: Re-encryption preserves the message point M . Bob can decrypt the new ciphertext with his own secret key and recover the same M that was in Alice's original ciphertext.

```

# Alice decrypts her own credential
M_decrypted = add(C_alice, neg(mul(R_alice, sk_alice))) # C - sk*R = M
assert eq(M_decrypted, M_alice), "Self-decryption failed"

# Re-encrypt M under Bob's public key with fresh randomness r'
r_prime = rand()
R_for_bob = mul(G1, r_prime) # R' = r' * G
C_for_bob = add(M_alice, mul(pk_bob, r_prime)) # C' = M + r' * pk_bob
E_for_bob = (R_for_bob, C_for_bob)

```

```

print("Alice re-encrypts her identity for Bob:")
print(f"  E_bob = (R', C') where:")
print(f"    R' = r'*G      = {pt(R_for_bob)}")
print(f"    C' = M+r'*pk    = {pt(C_for_bob)}")

# Verify Bob can decrypt
M_bob_decrypts = add(C_for_bob, neg(mul(R_for_bob, sk_bob)))
assert eq(M_bob_decrypts, M_alice), "Bob cannot decrypt E_bob"
print(f"\nBob decrypts: C' - sk_bob*R' = {pt(M_bob_decrypts)}")
print(f"Matches Alice's M:                {pt(M_alice)}")

```

```

Alice re-encrypts her identity for Bob:
  E_bob = (R', C') where:
    R' = r'*G      = (208313..., 777308...)
    C' = M+r'*pk    = (317012..., 982074...)

Bob decrypts: C' - sk_bob*R' = (739132..., 937433...)
Matches Alice's M:          (739132..., 937433...)

```

How to read the output: The last two lines show Bob's decrypted point and Alice's original M . The coordinate fragments must be identical – same point, different ciphertext wrapping it.

8.2 Chaum-Pedersen Proof of Correct Re-Encryption

Alice proves that E_{bob} encrypts the same message point M as her registered E_{alice} – without revealing m or any private key.

The proof demonstrates knowledge of the re-encryption randomness r' and the difference between the two ElGamal randomness values, such that both ciphertexts decrypt to the same M .

```

# Chaum-Pedersen proof that E_alice and E_bob encrypt the same M.
#
# The key relation: C_alice - sk_alice * R_alice = M = C_bob - r' * pk_bob
# Equivalently: C_alice - sk_alice * R_alice = C_bob - r' * pk_bob
#
# Alice proves she knows (sk_alice, r') such that:
#   sk_alice * R_alice = C_alice - M   (she can decrypt E_alice)
#   r' * pk_bob       = C_bob - M     (she created E_bob correctly)
#   r' * G             = R_bob        (consistent randomness)

# Commit: random blinders
k1 = rand() # blinder for sk_alice
k2 = rand() # blinder for r'

T1 = mul(R_alice, k1) # k1 * R_alice
T2 = mul(pk_bob, k2) # k2 * pk_bob
T3 = mul(G1, k2)     # k2 * G

# Fiat-Shamir challenge (binds all public inputs + Ethereum context)
e_cp = keccak(
    R_alice[0], R_alice[1], C_alice[0], C_alice[1], # E_alice
    R_for_bob[0], R_for_bob[1], C_for_bob[0], C_for_bob[1], # E_bob
    pk_alice[0], pk_alice[1], # pk_alice
    pk_bob[0], pk_bob[1], # pk_bob
    T1[0], T1[1], T2[0], T2[1], T3[0], T3[1], # commitments
    42, 1337, # msg.sender, spender (stand-ins)
)

```

```

)

# Responses
s1_cp = (k1 + e_cp * sk_alice) % ORDER
s2_cp = (k2 + e_cp * r_prime) % ORDER

proof_cp = (e_cp, s1_cp, s2_cp)
print("Chaum-Pedersen proof:")
print(f" e = {e_cp}")
print(f" s1 = {s1_cp}")
print(f" s2 = {s2_cp}")

Chaum-Pedersen proof:
e = 13092059294237547034072599863131217155980548767073273130399966767289281925403
s1 = 10803564579787626590175999387726963061205040271894122614356287626443109222640
s2 = 18498728416392382863393504023291687099816832383645912351288068359299568365837

```

8.3 Verify Chaum-Pedersen Proof (What the Contract Computes)

The BUCK contract reads E_{alice} from the IdentityRegistry (never from calldata), then verifies the proof. Cost: ~29,000 gas.

What this proves: E_{alice} and E_{bob} encrypt the same message point M – without revealing M , sk_{alice} , or the re-encryption randomness r' . Three checks:

- **Check 1:** r' is consistent with R_{bob}
- **Check 2:** Both ciphertexts decrypt to the same M
- **Check 3:** Fiat-Shamir challenge was honestly derived

```

# === VERIFIER (BUCK contract, on-chain, ~29K gas) ===
e_v, s1_v, s2_v = proof_cp

# The verifier knows: E_alice (from registry), E_bob (from calldata),
# pk_alice (from registry), pk_bob (from registry).

# Reconstruct commitments from responses and challenge:
# T1' = s1 * R_alice - e * (C_alice - C_bob + r'*pk_bob - sk*R_alice)
# Simplified verification equations:
# s1 * R_alice == T1 + e * sk_alice * R_alice
# == T1 + e * (C_alice - M)
# But verifier doesn't know M. Use the relation C_alice - sk*R = C_bob - r'*pk_bob = M:
# s1 * R_alice - s2 * pk_bob == T1 - T2 + e*(C_alice - C_bob)
# And: s2 * G == T3 + e * R_bob

# Check 1: s2 * G == T3 + e * R_for_bob
lhs1 = mul(G1, s2_v)
rhs1 = add(T3, mul(R_for_bob, e_v))
assert eq(lhs1, rhs1), "Chaum-Pedersen check 1 FAILED"
print("Check 1 passed: s2*G == T3 + e*R_bob")
print(" (Proves r' is consistent with R_bob)")

# Check 2: s1*R_alice - s2*pk_bob == (T1 - T2) + e*(C_alice - C_bob)
lhs2 = add(mul(R_alice, s1_v), neg(mul(pk_bob, s2_v)))
rhs2 = add(add(T1, neg(T2)), mul(add(C_alice, neg(C_for_bob)), e_v))
assert eq(lhs2, rhs2), "Chaum-Pedersen check 2 FAILED"
print("Check 2 passed: s1*R_a - s2*pk_b == (T1-T2) + e*(C_a - C_b)")

```

```

print(" (Proves both ciphertexts encrypt the same M)")

# Check 3: Fiat-Shamir challenge
e_recon = keccak(
    R_alice[0], R_alice[1], C_alice[0], C_alice[1],
    R_for_bob[0], R_for_bob[1], C_for_bob[0], C_for_bob[1],
    pk_alice[0], pk_alice[1],
    pk_bob[0], pk_bob[1],
    T1[0], T1[1], T2[0], T2[1], T3[0], T3[1],
    42, 1337,
)
assert e_recon == e_v, "Fiat-Shamir challenge mismatch"
print("Check 3 passed: Fiat-Shamir challenge verified")

print("\n==> approve() PASSED. Alice's re-encrypted identity stored for Bob.")
print("    _receiptFragments[alice][bob] = keccak256(E_bob)")

Check 1 passed: s2*G == T3 + e*R_bob
    (Proves r' is consistent with R_bob)
Check 2 passed: s1*R_a - s2*pk_b == (T1-T2) + e*(C_a - C_b)
    (Proves both ciphertexts encrypt the same M)
Check 3 passed: Fiat-Shamir challenge verified

==> approve() PASSED. Alice's re-encrypted identity stored for Bob.
    _receiptFragments[alice][bob] = keccak256(E_bob)

```

How to read the output: All three "passed" lines must appear. If Alice re-encrypted for a *different* public key or used a different M , Check 2 would fail. A counter-example follows.

8.3.1 Counter-example: Re-Encryption with Wrong Message

If Alice tries to pass off a different identity point in the re-encrypted ciphertext, Check 2 catches the discrepancy:

```

# Alice tries to re-encrypt a DIFFERENT M (e.g., someone else's identity)
m_other = rand()
M_other = mul(G1, m_other)
r_bad = rand()
R_bad = mul(G1, r_bad)
C_bad = add(M_other, mul(pk_bob, r_bad)) # encrypts M_other, not M_alice

# Build Chaum-Pedersen proof (Alice tries to cheat)
k1b = rand(); k2b = rand()
T1b = mul(R_alice, k1b); T2b = mul(pk_bob, k2b); T3b = mul(G1, k2b)
e_bad = keccak(
    R_alice[0], R_alice[1], C_alice[0], C_alice[1],
    R_bad[0], R_bad[1], C_bad[0], C_bad[1],
    pk_alice[0], pk_alice[1], pk_bob[0], pk_bob[1],
    T1b[0], T1b[1], T2b[0], T2b[1], T3b[0], T3b[1], 42, 1337)
s1b = (k1b + e_bad * sk_alice) % ORDER
s2b = (k2b + e_bad * r_bad) % ORDER

# Check 2: s1*R_a - s2*pk_b == (T1-T2) + e*(C_alice - C_bad)?
lhs = add(mul(R_alice, s1b), neg(mul(pk_bob, s2b)))
rhs = add(add(T1b, neg(T2b)), mul(add(C_alice, neg(C_bad)), e_bad))
ok = eq(lhs, rhs)
print(f"Forged re-encryption -- Check 2: {ok}")
print(f" (FAILS: C_bad encrypts a different M than C_alice.)")
print(f" (The contract rejects the approve() call.)")

```

```

Forged re-encryption -- Check 2: False
  (FAILS: C_bad encrypts a different M than C_alice.)
  (The contract rejects the approve() call.)

```

9 Transfer – Bilateral Identity Check

With Alice’s `approve` completed and Bob registered as a public account, the `transfer` can proceed. The contract performs a bilateral identity check: for each party, it verifies that either `isPublic` is true or a valid `_receiptFragments` entry exists from a prior `approve`. No cryptography runs at transfer time – only boolean flag checks. We simulate the check for our Private-to-Public scenario and show that both sides pass.

```

# Simulating the contract's bilateral identity check:
is_public_alice = False
is_public_bob   = True
receipt_alice_to_bob = True # set during approve() above
receipt_bob_to_alice = False # Bob never called approve (he's public)

# Sender (Alice) identity available to receiver (Bob)?
sender_ok = is_public_alice or receipt_alice_to_bob
# Receiver (Bob) identity available to sender (Alice)?
receiver_ok = is_public_bob or receipt_bob_to_alice

print("Bilateral identity check:")
print(f" Alice (private): isPublic={is_public_alice}, "
      f"receiptFragment={receipt_alice_to_bob} => {sender_ok}")
print(f" Bob   (public): isPublic={is_public_bob}, "
      f"receiptFragment={receipt_bob_to_alice} => {receiver_ok}")
print(f"\n sender_ok AND receiver_ok = {sender_ok and receiver_ok}")
print(f"\n==> transfer(bob, 500 BUCK) SUCCEEDS")
print(f"    BuckTransferReceipt emitted with identity hashes")

Bilateral identity check:
  Alice (private): isPublic=False, receiptFragment=True => True
  Bob   (public): isPublic=True, receiptFragment=False => True

sender_ok AND receiver_ok = True

==> transfer(bob, 500 BUCK) SUCCEEDS
    BuckTransferReceipt emitted with identity hashes

```

10 Bob Verifies Alice’s Identity (Two-Channel Design)

This is the two-channel design in action. Channel 1 (Ethereum): Bob decrypts E_{pool} with his secret key to recover the message point M . Channel 2 (Holochain): Bob receives Alice’s `identity_data` and verifies the binding $H(\text{identity_data}) \cdot G = M$. Neither channel alone is sufficient – the Ethereum ciphertext proves *which* identity was registered, while the Holochain data provides the *human-readable content*. A counter-example shows that changing even one character of `identity_data` in transit (e.g., "Alice" to "Mallory") produces a completely different point that doesn’t match M , so Bob detects tampering immediately.

```

# === Bob's side ===

# Channel 1 (Ethereum proof chain): Bob decrypts E_bob
M_from_bob = add(C_for_bob, neg(mul(R_for_bob, sk_bob)))
print("Bob decrypts E_bob from on-chain IdentityExchange event:")
print(f" M = C' - sk_bob * R' = {pt(M_from_bob)}")

# Channel 2 (Holochain data channel): Bob receives identity_data
received_identity_data = alice_identity_data # <-- via Holochain, encrypted for Bob
print(f"\nBob receives identity_data via Holochain:")
print(f" '{received_identity_data[:60]}...'")

# Verify binding: H(identity_data) * G == M
m_check = keccak(received_identity_data)
M_check = mul(G1, m_check)
assert eq(M_check, M_from_bob), "Identity binding FAILED"

print(f"\nTwo-channel verification:")
print(f" m = H(identity_data) = {m_check}")
print(f" m * G = {pt(M_check)}")
print(f" M      = {pt(M_from_bob)}")
print(f" H(identity_data) * G == M : True")
print(f"\n==> Bob has cryptographic proof that the identity_data")
print(f"      he received is the same identity the issuer certified")
print(f"      and Alice registered on-chain.")

Bob decrypts E_bob from on-chain IdentityExchange event:
M = C' - sk_bob * R' = (739132..., 937433...)

Bob receives identity_data via Holochain:
'{"date_of_birth":"1992-03-15","epoch":42,"family_name":"John...'}

Two-channel verification:
m = H(identity_data) = 6446313962173165014188361468977035479170665192834402473818305319846425860127
m * G = (739132..., 937433...)
M      = (739132..., 937433...)
H(identity_data) * G == M : True

==> Bob has cryptographic proof that the identity_data
      he received is the same identity the issuer certified
      and Alice registered on-chain.

```

How to read the output: The coordinate fragments of $m * G$ and M (decrypted from on-chain) must be identical. A counter-example follows showing what Bob sees if the Holochain data is tampered.

10.1 Counter-example: Tampered Identity Data

If a man-in-the-middle alters the identity data in transit, Bob detects it immediately – the hash no longer matches the on-chain point:

```

# Someone tampers with the identity data in transit
tampered = alice_identity_data.replace("Alice", "Mallory")
m_tampered = keccak(tampered)
M_tampered = mul(G1, m_tampered)

print(f"Tampered identity_data: ...given_name: 'Mallory'...")
print(f" H(tampered) * G = {pt(M_tampered)}")

```

```

print(f" M (from chain) = {pt(M_from_bob)}")
print(f" Match: {eq(M_tampered, M_from_bob)}")
print(f" (Bob rejects: the off-chain data doesn't match the on-chain point.)")

```

```

Tampered identity_data: ...given_name: 'Mallory'...
H(tampered) * G = (722726..., 633711...)
M (from chain) = (739132..., 937433...)
Match: False
(Bob rejects: the off-chain data doesn't match the on-chain point.)

```

11 Receipt Construction

After the transfer, both parties hold each other's plaintext `identity_data` (Alice has Bob's from the public IdentityRegistry; Bob has Alice's from Holochain) and the transfer metadata from on-chain events. Either party can independently construct a canonical receipt – deterministic JSON with sorted keys and no whitespace – and compute its `keccak256` hash. Alice's and Bob's independently constructed receipts produce identical hashes without coordination. This canonical form is what makes dispute resolution possible: both parties hold the same evidence, and neither can alter their copy without breaking the hash.

```

# Both parties construct the same canonical receipt
def canonical_receipt(sender_id, receiver_id, amount, tx_hash, block):
    return json.dumps({
        "sender": json.loads(sender_id),
        "receiver": json.loads(receiver_id),
        "amount": amount,
        "tx_hash": tx_hash,
        "block": block,
    }, sort_keys=True, separators=(',', ':'))

# Stand-in transaction data (would come from on-chain Transfer event)
tx_hash = "0xabcdef1234567890"
block = 19400000
amount = 500

# Alice constructs her copy
receipt_alice = canonical_receipt(
    alice_identity_data, bob_identity_data, amount, tx_hash, block)

# Bob constructs his copy
receipt_bob = canonical_receipt(
    alice_identity_data, bob_identity_data, amount, tx_hash, block)

h_alice = keccak(receipt_alice)
h_bob = keccak(receipt_bob)

print("Receipt construction (independent, canonical):")
print(f" Alice's H(receipt) = {h_alice}")
print(f" Bob's H(receipt) = {h_bob}")
print(f" Match: {h_alice == h_bob}")
print(f"\nReceipt stored (encrypted) on each party's Holochain source chain.")
print(f"The receipt hash anchors the off-chain record to the on-chain transfer.")

```

Receipt construction (independent, canonical):

```
Alice's H(receipt) = 21535858141339728776948445993719093154923735092821067258494283187515610795617
Bob's   H(receipt) = 21535858141339728776948445993719093154923735092821067258494283187515610795617
Match: True
```

Receipt stored (encrypted) on each party's Holochain source chain.

The receipt hash anchors the off-chain record to the on-chain transfer.

12 Summary: What Each Party Knows

After a complete Private-to-Public transfer, the information landscape is sharply partitioned. Eve (a passive on-chain observer) learns nothing about Alice's identity from any public artifact. Bob learns Alice's identity only because Alice explicitly authorized it via `approve`. The issuer can identify Alice given M , but cannot link Alice's accounts from on-chain data alone. No party – not even the issuer – can link Alice's rerandomized signature σ' back to the original σ .

```
[
  ["Artifact", "Alice", "Bob", "Eve", "Issuer"],
  None,
  ["identity_data (Alice)", "YES (owns it)", "YES (Holochain)", "NO", "YES (issued it)"],
  ["identity_data (Bob)", "YES (public)", "YES (owns it)", "YES (public)", "YES (issued it)"],
  ["m = H(identity_data)", "YES", "YES (computes)", "NO", "YES"],
  ["M = m*G", "YES", "YES (decrypts)", "NO", "NO"],
  ["E_alice (registered)", "YES", "sees on-chain", "sees on-chain", "NO"],
  ["E_bob (re-encrypted)", "YES (created)", "YES (decrypts)", "sees on-chain", "NO"],
  ["sk_alice", "YES", "NO", "NO", "NO"],
  ["sk_bob", "NO", "YES", "NO", "NO"],
  ["PS signature sigma", "YES", "NO", "NO", "YES (created)"],
  ["sigma' (rerandomized)", "YES", "sees on-chain", "sees on-chain", "UNLINKABLE"],
  ["Chaum-Pedersen proof", "YES (created)", "verified on-chain", "sees on-chain", "N/A"],
  ["Full receipt", "YES", "YES", "NO", "NO"],
  ["Link sigma to sigma'", "NO (stat. unlinkable)", "NO", "NO", "NO"],
]
```

Artifact	Alice	Bob	Eve	Issuer
identity_data (Alice)	YES (owns it)	YES (Holochain)	NO	YES (issued it)
identity_data (Bob)	YES (public)	YES (owns it)	YES (public)	YES (issued it)
m = H(identity_data)	YES	YES (computes)	NO	YES
M = m*G	YES	YES (decrypts)	NO	NO
E_alice (registered)	YES	sees on-chain	sees on-chain	NO
E_bob (re-encrypted)	YES (created)	YES (decrypts)	sees on-chain	NO
sk_alice	YES	NO	NO	NO
sk_bob	NO	YES	NO	NO
PS signature sigma	YES	NO	NO	YES (created)
sigma' (rerandomized)	YES	sees on-chain	sees on-chain	UNLINKABLE
Chaum-Pedersen proof	YES (created)	verified on-chain	sees on-chain	N/A
Full receipt	YES	YES	NO	NO
Link sigma to sigma'	NO (stat. unlinkable)	NO	NO	NO

13 Appendix: The Identity Fountain – Unlinkability Demonstration

Alice derives a second credential from the same PS signature – simulating a "savings" account alongside her "checking" account. We verify that the second credential is a valid PS signature on the same m , then compare all public artifacts (rerandomized signature, public key, ElGamal ciphertext) between the two accounts. Every coordinate fragment is completely different: no shared structure, no correlation. This is *statistical* unlinkability (information-theoretic, not merely computational) – meaning no algorithm, regardless of computing power, can determine that these two accounts belong to the same person from on-chain data alone. Both credentials verify under the same issuer public key, confirming they represent certified identities, but the issuer itself cannot link them without access to a counterparty's decrypted M .

```
# === Alice creates a SECOND account (e.g., "savings") ===
# Same m, same sigma, but fresh rerandomization and fresh keys

t_rerand_2 = rand()
sigma_p2_1 = mul(sigma[0], t_rerand_2)
sigma_p2_2 = mul(sigma[1], t_rerand_2)
sigma_p2    = (sigma_p2_1, sigma_p2_2)

sk_alice_2 = rand()
pk_alice_2 = mul(G1, sk_alice_2)

r_enc_2    = rand()
R_alice_2  = mul(G1, r_enc_2)
C_alice_2  = add(M_alice, mul(pk_alice_2, r_enc_2))
E_alice_2  = (R_alice_2, C_alice_2)

# Verify: sigma_p2 is a valid PS signature on the same m
lhs2 = pairing(add(IPK_X, mul(IPK_Y, m_alice)), sigma_p2[0])
rhs2 = pairing(G2, sigma_p2[1])
assert lhs2 == rhs2, "Second credential PS check FAILED"

print("Second credential derived (same identity, fresh account):")
print(f"  pk_alice_2 = {pt(pk_alice_2)}")
print(f"  E_alice_2  = (R, C) -- independent ciphertext")
print(f"  sigma'_2   = {pt(sigma_p2[0])}")
print()
print("Unlinkability check:")
print(f"  sigma'_1 (acct 1) = {pt(sigma_p[0])}")
print(f"  sigma'_1 (acct 2) = {pt(sigma_p2[0])}")
print(f"  pk (acct 1)       = {pt(pk_alice)}")
print(f"  pk (acct 2)       = {pt(pk_alice_2)}")
print(f"  E.R (acct 1)      = {pt(R_alice)}")
print(f"  E.R (acct 2)      = {pt(R_alice_2)}")
print()
print("  All values are independent.  No shared structure.")
print("  PS rerandomization is STATISTICALLY unlinkable --")
print("  not merely computationally hard, but information-")
print("  theoretically impossible to correlate.")
print()
print("  Both credentials verify under the same issuer public key,")
print("  but zero bits of mutual information exist between them.")
```

Second credential derived (same identity, fresh account):

```
pk_alice_2 = (960817..., 598752...)
E_alice_2  = (R, C) -- independent ciphertext
sigma'_2   = (918313..., 1359...)
```

Unlinkability check:

```
sigma'_1 (acct 1) = (277377..., 422277...)
sigma'_1 (acct 2) = (918313..., 1359...)
pk (acct 1)      = (561696..., 359461...)
pk (acct 2)      = (960817..., 598752...)
E.R (acct 1)     = (778258..., 705687...)
E.R (acct 2)     = (186360..., 959644...)
```

All values are independent. No shared structure.
 PS rerandomization is STATISTICALLY unlinkable --
 not merely computationally hard, but information-
 theoretically impossible to correlate.

Both credentials verify under the same issuer public key,
 but zero bits of mutual information exist between them.

How to read the output: Compare the coordinate fragments between "acct 1" and "acct 2" for σ'_1 , pk , and $E.R$. They must all be completely different (random-looking). If any pair matched, the unlinkability property would be broken.

14 Appendix: Small-Group Proofs

The BN254 examples above prove the cryptography at production scale, but 77-digit numbers resist hand-verification. These demonstrations use modulus $p = 23$ working in an order-11 subgroup where every value fits in two digits.

EC (additive)	Modular (multiplicative)
Scalar mult $m \cdot G$	Exponentiation $g^m \bmod p$
Point addition $P + Q$	Multiplication $P \cdot Q \bmod p$

Schnorr proofs require prime-order groups, so we use $g = 4$, which generates the order-11 subgroup (the quadratic residues mod 23). All exponent arithmetic is mod $q = 11$.

14.1 Setup and ElGamal

ElGamal encryption of a group element $M = g^m$ under public key $pk = g^{sk}$ with randomness r :

$$E = (R, C) = (g^r, M \cdot pk^r) \pmod{p}$$

$$\text{Decrypt: } M = C \cdot (R^{sk})^{-1} \pmod{p}$$

```
import hashlib

p = 23; q = 11; g = 4      # safe prime, prime-order subgroup

def mod_exp(base, exp, mod):
    result = 1
```

```

    base = base % mod
    while exp > 0:
        if exp & 1:
            result = (result * base) % mod
            base = (base * base) % mod
            exp >>= 1
    return result

def mod_inv(a, mod):
    def egcd(a, b):
        if a == 0: return b, 0, 1
        g, x, y = egcd(b % a, a)
        return g, y - (b // a) * x, x
    _, x, _ = egcd(a % mod, mod)
    return x % mod

def fs_hash(*args):
    """Fiat-Shamir: H(args...) mod q"""
    h = hashlib.sha256()
    for a in args:
        h.update(str(a).encode())
    return int(h.hexdigest(), 16) % q

# Identity
m = 3; M = mod_exp(g, m, p)

# Keys
sk_a = 5; pk_a = mod_exp(g, sk_a, p)
sk_b = 7; pk_b = mod_exp(g, sk_b, p)

# Encrypt for Alice (r_a=2) and Bob (r_b=9)
r_a = 2; R_a = mod_exp(g, r_a, p)
C_a = (M * mod_exp(pk_a, r_a, p)) % p

r_b = 9; R_b = mod_exp(g, r_b, p)
C_b = (M * mod_exp(pk_b, r_b, p)) % p

# Decrypt both
dec_a = (C_a * mod_inv(mod_exp(R_a, sk_a, p), p)) % p
dec_b = (C_b * mod_inv(mod_exp(R_b, sk_b, p), p)) % p

[
    ["", "m", "M=g^m", "sk", "pk", "r", "R=g^r", "C=M*pk^r", "Decrypted"],
    None,
    ["Alice", m, M, sk_a, pk_a, r_a, R_a, C_a, dec_a],
    ["Bob", m, M, sk_b, pk_b, r_b, R_b, C_b, dec_b],
]

```

	m	M=g ^m	sk	pk	r	R=g ^r	C=M*pk ^r	Decrypted
Alice	3	18	5	12	2	16	16	18
Bob	3	18	7	8	9	13	1	18

Same message $M = 18$ in both ciphertexts: (16, 16) for Alice, (13, 1) for Bob. Different randomness makes them unlinkable.

14.2 NIZK: Proof of Knowledge (Registration)

At registration, Alice proves she knows the plaintext m and randomness r inside her ElGamal ciphertext – without revealing either. (In the full protocol, a Pointcheval-Sanders pairing check additionally binds m to the issuer's signature; this demonstration isolates the sigma-protocol core.)

Statement: $R = g^r, \quad C = g^m \cdot pk^r$

Commit: $T_R = g^{k_r}, \quad T_C = g^{k_m} \cdot pk^{k_r}$

Challenge: $e = H(g, pk, R, C, T_R, T_C) \bmod q$

Respond: $s_m = k_m - e \cdot m \pmod{q}, \quad s_r = k_r - e \cdot r \pmod{q}$

Verify 1: $g^{s_r} \cdot R^e \stackrel{?}{=} T_R$

Verify 2: $g^{s_m} \cdot pk^{s_r} \cdot C^e \stackrel{?}{=} T_C$

```
# Prove knowledge of (m=3, r=2) in E_alice = (16, 16)
k_m = 1; k_r = 1                                     # random nonces
T_R = mod_exp(g, k_r, p)                             # 4
T_C = (mod_exp(g, k_m, p) * mod_exp(pk_a, k_r, p)) % p # 2

e = fs_hash(g, pk_a, R_a, C_a, T_R, T_C)             # 8

s_m = (k_m - e * m) % q                               # 10
s_r = (k_r - e * r_a) % q                             # 7

chk1 = (mod_exp(g, s_r, p) * mod_exp(R_a, e, p)) % p
chk2 = (mod_exp(g, s_m, p) * mod_exp(pk_a, s_r, p) * mod_exp(C_a, e, p)) % p

print(f"Commit:   T_R = {T_R}, T_C = {T_C}")
print(f"Challenge: e = {e}")
print(f"Respond:   s_m = {s_m}, s_r = {s_r}")
print(f"Verify 1:  g^{s_r} * R^e = {chk1:2} == T_R={T_R:2} {'PASS' if chk1 == T_R else 'FAIL'}")
print(f"Verify 2:  g^{s_m} * pk^{s_r} * C^e = {chk2:2} == T_C={T_C:2} {'PASS' if chk2 == T_C else 'FAIL'}")

# Counter-example: Alice claims m'=5 (wrong identity)
m_fake = 5
s_m_fake = (k_m - e * m_fake) % q
chk2_f = (mod_exp(g, s_m_fake, p) * mod_exp(pk_a, s_r, p) * mod_exp(C_a, e, p)) % p
print(f"\nCounter-example (m'={m_fake}):")
print(f" Verify 2:  {chk2_f} == T_C={T_C} {'PASS' if chk2_f == T_C else 'FAIL'}")
print(f" The ciphertext encrypts m={m}, not m'={m_fake}.")
print(f" No response value can bridge the gap.")

Commit:   T_R = 4, T_C = 2
Challenge: e = 8
Respond:   s_m = 10, s_r = 7
Verify 1:  g^{s_r} * R^e = 4 == T_R= 4 PASS
Verify 2:  g^{s_m} * pk^{s_r} * C^e = 2 == T_C= 2 PASS

Counter-example (m'=5):
Verify 2:  4 == T_C=2 FAIL
The ciphertext encrypts m=3, not m'=5.
No response value can bridge the gap.
```

How it works: The prover commits to random nonces k_m, k_r before seeing the challenge e . Only someone who knows the real m and r can produce responses that satisfy both equations for the specific e the Fiat-Shamir hash generates. Check 1 confirms knowledge of r ; Check 2 confirms knowledge of m and its consistency with the ciphertext. A wrong m' produces a wrong s_m , which ripples through the multi-exponentiation in Check 2.

14.3 Chaum-Pedersen: Proof of Same Plaintext (Approve)

At **approve** time, Alice re-encrypts her identity for Bob and proves both ciphertexts contain the same M – without revealing it. Three independent checks verify her key ownership, her knowledge of the new randomness, and the binding constraint that the decrypted values match.

$$\text{Statement: } C_a \cdot (R_a^{sk_a})^{-1} = C_b \cdot (pk_b^{r_b})^{-1} \quad (\text{same } M)$$

$$\text{Commit: } T_1 = g^{k_1}, \quad T_2 = g^{k_2}, \quad T_3 = pk_b^{k_2} \cdot R_a^{-k_1}$$

$$\text{Challenge: } e = H(g, pk_a, pk_b, R_a, C_a, R_b, C_b, T_1, T_2, T_3)$$

$$\text{Respond: } s_1 = k_1 - e \cdot sk_a, \quad s_2 = k_2 - e \cdot r_b \quad (\text{mod } q)$$

$$\text{Verify 1: } g^{s_1} \cdot pk_a^e \stackrel{?}{=} T_1 \quad (\text{owns } pk_a)$$

$$\text{Verify 2: } g^{s_2} \cdot R_b^e \stackrel{?}{=} T_2 \quad (\text{knows } r_b)$$

$$\text{Verify 3: } pk_b^{s_2} \cdot R_a^{-s_1} \cdot (C_b/C_a)^e \stackrel{?}{=} T_3 \quad (\text{same } M)$$

```
# Prove E_alice=(16,16) and E_bob=(13,1) encrypt the same M
k1 = 1; k2 = 1
T1 = mod_exp(g, k1, p) # 4
T2 = mod_exp(g, k2, p) # 4
T3 = (mod_exp(pk_b, k2, p) * mod_exp(R_a, (q - k1) % q, p)) % p # 12

e_cp = fs_hash(g, pk_a, pk_b, R_a, C_a, R_b, C_b, T1, T2, T3) # 4

s1 = (k1 - e_cp * sk_a) % q # 3
s2 = (k2 - e_cp * r_b) % q # 9

v1 = (mod_exp(g, s1, p) * mod_exp(pk_a, e_cp, p)) % p
v2 = (mod_exp(g, s2, p) * mod_exp(R_b, e_cp, p)) % p
cb_ca = (C_b * mod_inv(C_a, p)) % p
v3 = (mod_exp(pk_b, s2, p) * mod_exp(R_a, (q - s1) % q, p)
      * mod_exp(cb_ca, e_cp, p)) % p

print(f"Commit: T1={T1}, T2={T2}, T3={T3}")
print(f"Challenge: e = {e_cp}")
print(f"Respond: s1={s1}, s2={s2}")
print(f"Verify 1 (owns pk_a): {v1:2} == T1={T1:2} {'PASS' if v1 == T1 else 'FAIL'}")
print(f"Verify 2 (knows r_b): {v2:2} == T2={T2:2} {'PASS' if v2 == T2 else 'FAIL'}")
print(f"Verify 3 (same M): {v3:2} == T3={T3:2} {'PASS' if v3 == T3 else 'FAIL'}")

# Counter-example: Alice encrypts M'=g^7=8 instead of M=g^3=18
m_bad = 7; M_bad = mod_exp(g, m_bad, p)
C_b_bad = (M_bad * mod_exp(pk_b, r_b, p)) % p
```

```

e_bad = fs_hash(g, pk_a, pk_b, R_a, C_a, R_b, C_b_bad, T1, T2, T3)
s1_b = (k1 - e_bad * sk_a) % q
s2_b = (k2 - e_bad * r_b) % q
v1_b = (mod_exp(g, s1_b, p) * mod_exp(pk_a, e_bad, p)) % p
v2_b = (mod_exp(g, s2_b, p) * mod_exp(R_b, e_bad, p)) % p
cb_ca_b = (C_b_bad * mod_inv(C_a, p)) % p
v3_b = (mod_exp(pk_b, s2_b, p) * mod_exp(R_a, (q - s1_b) % q, p)
        * mod_exp(cb_ca_b, e_bad, p)) % p

print(f"\nCounter-example (M'=g^{m_bad}={M_bad}, C_b'={C_b_bad}):")
print(f" Verify 1 (owns pk_a): {v1_b:2} == T1={T1:2} {'PASS' if v1_b == T1 else 'FAIL'}")
print(f" Verify 2 (knows r_b): {v2_b:2} == T2={T2:2} {'PASS' if v2_b == T2 else 'FAIL'}")
print(f" Verify 3 (same M): {v3_b:2} == T3={T3:2} {'PASS' if v3_b == T3 else 'FAIL'}")
print(f" Alice owns her key and knows r_b -- checks 1,2 pass.")
print(f" But E_alice encrypts M={M} while E_bob encrypts M'={M_bad}.")

```

```

Commit:    T1=4, T2=4, T3=12
Challenge: e = 4
Respond:   s1=3, s2=9
Verify 1 (owns pk_a):    4 == T1= 4  PASS
Verify 2 (knows r_b):    4 == T2= 4  PASS
Verify 3 (same M):      12 == T3=12  PASS

```

```

Counter-example (M'=g^7=8, C_b'=3):
Verify 1 (owns pk_a):    4 == T1= 4  PASS
Verify 2 (knows r_b):    4 == T2= 4  PASS
Verify 3 (same M):       1 == T3=12  FAIL
Alice owns her key and knows r_b -- checks 1,2 pass.
But E_alice encrypts M=18 while E_bob encrypts M'=8.

```

How it works: Check 3 is the binding constraint. Expanding the verification equation, it passes only when $C_a/R_a^{sk_a} = C_b/pk_b^{r_b}$ – i.e., decrypting E_{alice} with Alice's key yields the same M as de-randomizing E_{bob} with r_b . A different message breaks this equality, producing a mismatch that no choice of responses can hide, because the commitments were fixed before the challenge was known.